Evaluating Annual Sales Performance and Revenue Drivers for Vertex Electronics

Overview

This analysis evaluates the 2025 sales performance of Vertex Electronics, focusing on revenue, impact of discounts, product performance, and customer purchasing patterns. The objective was to identify key revenue drivers, top-performing products and regions, and patterns in customer behavior that affected profitability during 2025.

The findings highlight top-performing products, monthly revenue trends, regional performance patterns, category contributions, the relationship between quantity and revenue, and how discounts influenced sales performance. These historical insights provide guidance for optimizing pricing strategies, product offerings, and overall revenue growth in 2026..

Data Source

The data used for this analysis was obtained from a simulated retail sales dataset for Vertex Electronics, covering transactions over a one-year period (2025).

The dataset includes the following columns:

- **Order ID:** Unique identifier for each transaction.
- **Order Date:** Date the order was placed.
- **Month Name:** Month in which the order occurred.
- **Customer ID:** Unique identifier for each customer.
- **Region:** Customer's geographical location.
- **Product Category:** Type or category of the product.
- **Product Name:** Specific product purchased.
- **Quantity:** Number of units purchased.
- **Unit Price:** Price per single unit of the product.
- **Total Sales (Before Discount):** Total value of the order before discounts.
- **Discount:** Discount applied to the order.

Revenue After Discount: Actual revenue earned after discounts.

- **Payment Method:** Method used to make the payment.

Problem Statement

Vertex Electronics operated in a competitive retail environment in 2025. While detailed transaction-level sales data exists, there was limited visibility into how product performance, regional demand, pricing strategies, and order quantities influenced overall revenue during that year.

Historical insights are needed to help the business adjust its strategy for this year, 2026 by understanding which products, regions, and strategies were most effective. Without this understanding, it would be difficult to make informed decisions on product assortment, pricing, and revenue optimization.

Specific Business Problems Being Addressed

- Analyzing monthly sales trends in 2025 to identify patterns, seasonality, and performance fluctuations
- Identifying top-performing products and categories in 2025 that drove the majority of revenue
- Evaluating regional sales performance to determine historically high- and low-performing markets
- Assessing the impact of discounts on revenue and sales during 2025
- Understanding how order quantity influenced revenue generation
- Identifying opportunities to optimize product offerings, pricing strategies, and revenue growth for 2026

Success Criteria for Solving This Problem:

The analysis aims to generate actionable insights that will:

- Highlight key revenue drivers across products, regions, and time periods in 2025
- Identify top-performing products and categories for strategic focus in 2026
- Evaluate the effectiveness of discount strategies on sales performance
- Provide insights into customer purchasing patterns and order behavior
- Support data-driven decision-making for product strategy, pricing, and revenue optimization in 2026

Tools and Methodology

Tools

- Microsoft Excel: Dashboard creation, pivot tables, and visualization.
- Power Query: Data cleaning and transformation.
- Power Pivot: writing DAX measures for key metrics.

- SQL (PostgreSQL): Queried the sales dataset to analyze revenue, product performance, regional trends, and discount impact.

Methodology

- Data Cleaning: Inspected the sales dataset for inconsistencies, duplicates and missing values. The original dataset contained 1,250 records across 13 columns. During the data cleaning process, 50 duplicate records were identified and removed. Additionally, 194 records with missing values in key fields (Quantity and Total Sales) were excluded to ensure data accuracy and consistency. This resulted in a final dataset of 1,006 cleaned records used for analysis. I ensured correct data formats, and validated revenue calculations.
- Data Processing: Used SQL queries, Excel pivot tables and DAX measures in Power Pivot to calculate key metrics such as MoM%, total revenue, revenue after discount, average order value, and product/category contributions.
- Sales Analysis: Analyzed monthly revenue trends, product performance, regional sales patterns, and the impact of order quantity and discounts on revenue to identify key revenue drivers.
- Data Visualization: Built an interactive Excel dashboard to display key metrics, revenue trends, product performance, regional insights, and discount impacts, providing actionable insights for optimizing 2026 strategies.

Dashboard Overview

The dashboard presents a comprehensive view of Vertex Electronics' 2025 sales performance, highlighting revenue trends, product performance, regional insights, and the impact of discounts on profitability. It illustrates how order size influences revenue and identifies patterns that can guide pricing and product strategy decisions 2026.

Key Analysis Findings

Revenue Performance

- How did sales perform throughout 2025?

Vertex Electronics generated approximately $2.07M in total revenue after discounts across 1,006 orders, selling 2,979 products. Monthly revenue fluctuated throughout the year, reflecting changes in customer demand and seasonal trends. The highest revenue was recorded in October ($228,649), while the lowest occurred in July ($104,431).

These figures highlight periods of strong and weak sales performance, showing that revenue peaks during August and October, whereas mid-year sales were slower.

Month-over-Month Revenue Trends

- How did revenue change from month to month throughout 2025?

Monthly revenue showed significant volatility throughout 2025, with both sharp increases and steep declines recorded across the year. February opened positively with a 7% increase over January, followed by a strong March

surge of 20%, suggesting a mid-quarter demand spike. However, April saw a sharp reversal of -29%, indicating a notable pullback in customer spending.

Recovery came in May (+1%) and June (+14%), before July recorded the most dramatic decline of the year at -40%, bringing revenue to its lowest point of $104,431. This was followed by a remarkable August rebound of +109%, the strongest single-month growth of 2025, reflecting either a major promotional push or a surge in seasonal demand.

September again declined (-35%), but October recovered strongly (+61%), aligning with the year's peak revenue of $228,649. The final quarter showed a gradual slowdown, with November falling -20% and December closing the year at -11%.

Overall, 2025 revenue followed a volatile pattern rather than a steady growth trajectory, with alternating peaks and troughs throughout the year. The extreme swings particularly July's -40% drop and August's +109% rebound suggest that external factors such as promotions, stock availability, or seasonal demand played a significant role in monthly performance.

Key MOM% Highlights:

- Strongest growth month: August (+109%)
- Steepest decline month: July (-40%)
- Most stable period: January to March (gradual growth)
- Year-end trend: Gradual slowdown in Q4

Regional Performance

- How did sales perform across regions in 2025?

Sales performance differed significantly across regions, with the North region consistently generating the highest revenue. In the Electronics category, tablets led the sales with $322,079 in revenue. In Furniture, sofas contributed $209,379, while in Home Appliances, microwaves generated $222,348. By contrast, the East region recorded the lowest revenue, highlighting a regional gap in sales performance.

These results indicate that the North is a key market for high-value products, driving substantial portions of overall revenue.

Discount Impact

- How did discounts influence overall revenue in 2025?

Vertex Electronics applied discounts across products, resulting in an average Discount Impact of 9%. Despite these reductions, total revenue after discounts reached approximately $2.07M, showing that strategic discounting effectively drove sales without significantly eroding profitability.

This indicates that discounts can boost order volume while still preserving overall revenue

Quantity vs Revenue

- How does order size affect revenue in 2025?

Analysis of order quantities against revenue after discounts shows that higher-volume orders generally contributed more to total revenue, though some high-priced items sold in smaller quantities still generated significant revenue. For example, bulk purchases of tablets in Electronics and microwaves in Home Appliances drove substantial revenue, while premium items with lower quantities also maintained profitability.

This pattern suggests that both order volume and product price point are key drivers of revenue.

Payment Methods

- Which payment methods were most commonly used in 2025?

Orders were fairly evenly distributed across all payment methods, with Credit Card (345 orders), Bank Transfer (333 orders), and Cash (328 orders) all widely used.

This balance shows that customers appreciate having multiple convenient payment options, and no single method dominates overwhelmingly. Providing a variety of payment methods allows Vertex Electronics to accommodate different customer preferences, ensuring a smooth purchasing experience and supporting sales across all channels.

Limitations

- **Limited Historical Data:** The dataset covers only 2025, so trends may not reflect longer-term performance.
- **Customer Tracking Constraints:** Returning customers or repeat orders could not be fully analyzed due to missing or altered customer IDs.
- **Revenue and Profit Constraints:** Revenue calculations are based on sales after discounts, but cost prices, expenses, and profit margins were not included, limiting insights into overall profitability.

Recommendations

1. **Focus on High-Performing Regions**

The North region consistently generated the highest revenue across Electronics, Furniture, and Home Appliances. Vertex Electronics should prioritize inventory allocation, promotions, and marketing efforts in this region to maximize sales.

For the East region, which recorded the lowest revenue despite similar product availability, a dedicated regional sales strategy including localized pricing, targeted digital campaigns, and regional sales representatives should be explored for 2026.

2. **Optimize Product Strategy**

High-demand products like tablets, microwaves, and sofas drove substantial revenue. Maintain sufficient stock of these items and consider bundling or promotional strategies to further increase sales.

Monitor high-priced items with low quantity and low-priced items with high quantity to balance profitability and sales volume effectively.

3. **Leverage Discounts Strategically**

The average 9% discount successfully boosted sales without significantly reducing revenue. Discounts should continue to be targeted to key products and peak months to drive volume while maintaining profitability. However, discount thresholds should be reviewed quarterly to ensure long-term profitability is not gradually eroded.

4. **Maintain Multiple Payment Options**

Customers used Credit Card, Bank Transfer, and Cash almost equally. Continue offering all payment methods to accommodate customer preferences and support seamless transactions across all channels.

5. **Monitor Order Quantities**

Both high-volume and high-value orders contribute significantly to revenue. Planning inventory, promotions, and pricing strategies around these trends can help maximize overall sales performance.

6. **Plan for Seasonal Trends**

Peak revenue months were August and October, while July recorded the steepest decline. Month-over-month analysis revealed a -40% drop in July followed by a remarkable +109% rebound in August, suggesting that promotional activity and seasonal demand were key drivers. Planning targeted marketing campaigns and stock preparation specifically ahead of July can help minimize this dip and capitalize on the August surge in 2026.

7. **Customer Loyalty and Engagement**

Since returning or repeat customers could not be fully analyzed in 2025, implement better customer ID tracking for 2026. Use this data to design loyalty programs, personalized promotions, and retention strategies to increase repeat purchases and lifetime customer value.

Conclusion

The analysis of Vertex Electronics' 2025 sales performance highlights key drivers of revenue, including high-demand products, regional trends, order quantities, and the strategic use of discounts. Total revenue after discounts reached approximately $2.07M, with August and October as the strongest months, while the East region lagged behind other regions.

Month-over-month analysis further revealed significant revenue volatility throughout the year, with August recording the sharpest single-month growth at +109% and July the steepest decline at -40%, suggesting that seasonal demand and promotional activity were key influencers of monthly performance.

High-performing products like tablets, microwaves, and sofas contributed significantly to revenue, and discounts of around 9% helped boost order volume without severely impacting profitability. Payment methods were evenly distributed, showing the importance of offering multiple options to accommodate customer preferences.

It is important to note that returning or repeat customers could not be fully analyzed due to missing or altered customer IDs. This limits insights into customer loyalty and long-term purchasing patterns.

Overall, these findings provide clear guidance for 2026: focus on high-performing regions and products, maintain strategic discounting, plan inventory around seasonal trends, and ensure a variety of payment options. Implementing these strategies can maximize revenue, optimize operational planning, and strengthen overall business performance.

Links

Interact with the Dashboard
Clean Sales Dataset
Dirty Sales Dataset
SQL Queries